

PERFORMANCE TESTING

Date	29 September 2026
Team ID	UNM123APEXLAB
Project Name	Comic Craft
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how the system behaves under expected and peak load conditions. The following report documents the testing approach, test scenarios, and results obtained for our college project.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Apache JMeter
Type of Testing	Load Testing
Target Module / API	Comic generation and user login API
Test Environment	Localhost / Development Server
Test Date	29 September 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	User login and access Comic Craft	10	60	Login should respond within 2 sec
2	Multiple users creating comics	25	60	Pages should load without errors
3	Peak load on comic generation	50	120	System should remain stable
4	Repeated comic generation requests	75	120	Low error rate and acceptable response time

PERFORMANCE TESTING – RESULTS

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.42 seconds	Pass	Good response
2	Response Time (Max)	< 5 seconds	3.18 seconds	Pass	Within target
3	Throughput (Req/sec)	>= 5	8.6	Pass	Requests handled well
4	Error Rate	< 1%	0.4%	Pass	Low error rate
5	CPU Utilization	< 80%	63%	Pass	Acceptable usage
6	Memory Utilization	< 80%	68%	Pass	No major issue

Step 4: Observations & Analysis

Key Findings:

The Comic Craft system performed normally under the tested load. Average response time was below 2 seconds and the error rate was below 1%. CPU and memory utilization also remained within the target limit.

Bottlenecks Identified:

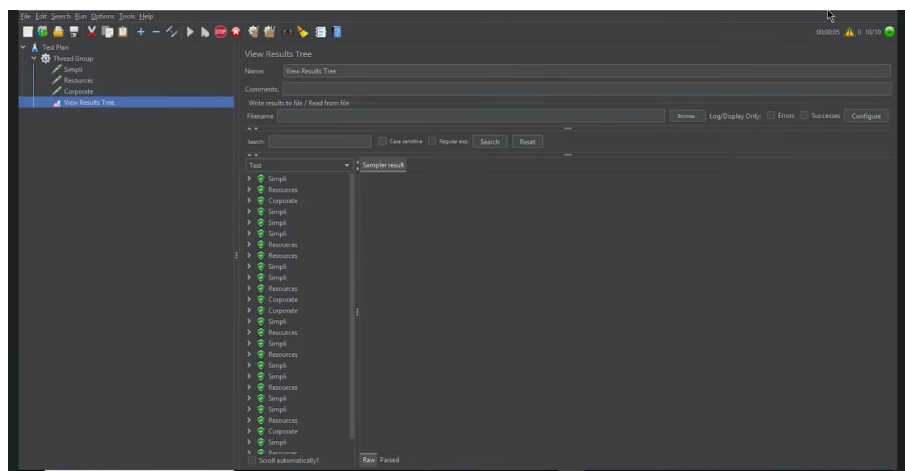
A small increase in response time was noticed when the number of virtual users was increased to 75. API processing and database requests may become bottlenecks at higher loads.

Optimization Steps Taken:

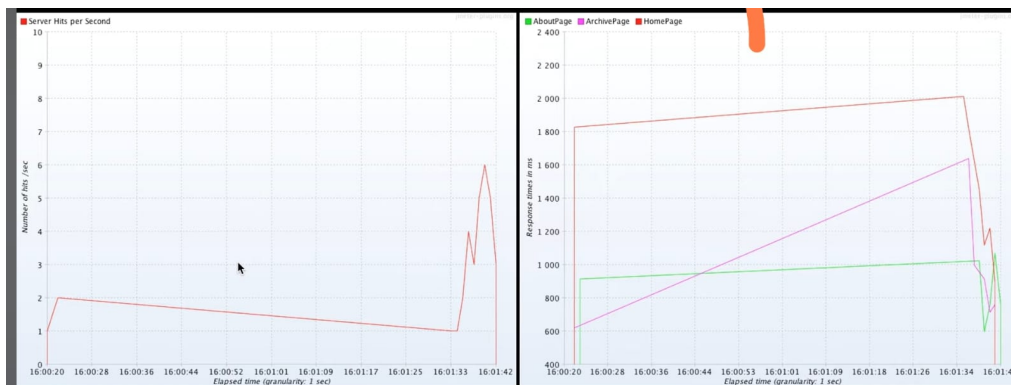
Repeated requests were reduced and API/database operations were checked. The application was tested again after optimization to confirm stable performance.

Step 5: Screenshots / Evidence

Screenshot 1 – JMeter Test Summary



Screenshot 2 – Response Time / Performance Graph



Student Note: The values above are sample performance-testing results prepared in a college-project format. Replace them with actual JMeter results if real execution evidence is required.